Bindu Madhava Koilada

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.74
Narayana junior college	Class XII (ssc)	2019	9.94%
Narayana school	Class X (ssc)	2017	10.00%

SCHOLASTIC ACHIEVEMENTS

Sunshine Organization : Honoured with star performer award for contributing in activities held in last academic year.

Winners in Innovation category : For Pneumatic brakes and clutch we got the Innovation award at SUPRA event held at buddh International Circuit , Greater Noida, in the month of June 2023 . Also we created a record of 2.5 sec in EGRESS test.

POSITIONS OF RESPONSIBILITY

Placement Coordinator of Additive Manufacturing program :

- Coordinating student profiles, recruiter communication and placement-related activities for the program.

Head of Event Coordination Team, IKS-SCh Club :

- Leading the event coordination team for planning and execution of student activities and institute-level events.

PG Buddy, Sunshine Organization :

- Supported incoming postgraduate students in academic, administrative and campus-related orientation.
- Assisted students in adapting to the academic and campus environment at IIT Hyderabad.

Class Representative of Additive Manufacturing program :

- Facilitated communication between students and faculty regarding academic and administrative activities.

Head of Chassis Department SAE club, NIT Kurukshetra :

- Led the design, development, and fabrication of the competition vehicle chassis for Team Accelerons in SUPRA.

SKILLS

- **CAD** : SolidWorks **Simulation**: ANSYS Fluent, Abaqus **Computational Tools:** MATLAB
- **Additive Manufacturing:** Materialise Magics, Simufact Additive, Ultimaker Cura, Bambu Studio.
- **Laser Processing:** Laser Welding, Laser Cladding, Laser Directed Energy Deposition, Laser Surface Processing.
- **Materials & Characterization:** Metallographic Sample Preparation , Microstructure Characterization, EBSD , Optical Microscopy, Hardness Testing.

RELEVANT COURSES

Mechanical Engineering : Strength of Materials , Machine Design, Material Science, Manufacturing Processes, Theory of Machines, Engineering Mechanics, Thermodynamics, Fluid Mechanics.

Additive Manufacturing : Fundamentals of Additive manufacturing, Materials for Additive manufacturing , Design for Metal additive manufacturing, Advanced material joining process, Finite Element Methods, Machine learning.

PROJECTS

Thesis Project : "Design and development of line beam based laser material processing head with thermal monitoring system."

- Designed a 16 mm × 2 mm line-beam laser processing head by converting a circular beam into a line beam profile.
- Integrated an IR pyrometer and in-line thermal imaging camera for real-time melt pool temperature acquisition.
- Developing analytical and numerical thermal models to correlate laser parameters,cooling rate, Microstructure evolution.

Course Project : "CFD analysis of mixed convection of air in a 2D heated backward-facing step using ANSYS fluent".

- Developed a 2D CFD model to investigate mixed convection flow and heat transfer in a heated backward-facing step.
- Performed computational analysis of velocity and temperature fields under mixed convection conditions using ANSYS .
- Studied flow behaviour and thermal characteristics associated with recirculation and heat transfer.

SAE CLUB : NIT Kurukshetra

- Designed and fabricated chassis for SUPRA 2022- 2023 for Team Accelerons.

B.Tech project : NIT Kurukshetra

- Design and simulation of V12 ENGINE using SOLID WORKS and thermal analysis in ANSYS software.

TRAINING AND WORKSHOP

Attended two-day workshop on additive manufacturing simulation using SIMUFACT software.

Attended three-day workshop on build and data preparation for 3D printing using MATERIALISE MAGICS software.

INDUSTRIAL VISITS

Azad Engineering Ltd : Observed the surface hardening of Steel Turbines using Robotic Arm with 2kw square beam laser .

Innomet Advanced Materials Ltd : Observed live production of Metal Powders and Tungsten Heavy Alloys .

EXTRACURRICULAR ACTIVITIES

- Completed 5K run in half marathon event held at SNCC of IIT Hyderabad in November 2025.
- **Languages :**Telugu(Native), Hindi(Proficient), English(Proficient), Japanese(Intermediate).
- **Sports :** Badminton ,Volley ball , Cricket.